

Library Management & Recommendation System

Database Schema Documentation & Data Dictionary

1. Table: Authors

Stores information about book authors.

Column	Data Type	Constraint	Description
author_id	INT	PRIMARY KEY, AUTO_INC	Unique identifier for each author.
author_name	VARCHAR(100)	NOT NULL	Full name of the author.

2. Table: Categories

Defines book genres and categories.

Column	Data Type	Constraint	Description
category_id	INT	PRIMARY KEY, AUTO_INC	Unique identifier for each category.
category_name	VARCHAR(50)	NOT NULL	Name of the genre (e.g., Sci-Fi, History).

3. Table: Books

The core inventory table for library assets.

Column	Data Type	Constraint	Description
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book_id	INT	PRIMARY KEY, AUTO_INC	Unique identifier for each book.
title	VARCHAR(150)	NOT NULL	The title of the book.
author_id	INT	FOREIGN KEY	References Authors(author_id).
category_id	INT	FOREIGN KEY	References Categories(category_id).
publication_year	INT	-	Year the book was published.
available_copies	INT	DEFAULT 5	Number of copies currently on shelf.

4. Table: Members

Contains details of registered library users.

Column	Data Type	Constraint	Description
member_id	INT	PRIMARY KEY, AUTO_INC	Unique identifier for each member.
member_name	VARCHAR(100)	NOT NULL	Full name of the member.
email	VARCHAR(100)	UNIQUE	Contact email address.
join_date	DATE	-	Date the membership was created.

5. Table: Borrow_Records

Tracks the movement of books (Transactions).

Column	Data Type	Constraint	Description
borrow_id	INT	PRIMARY KEY, AUTO_INC	Unique transaction ID.
member_id	INT	FOREIGN KEY	References Members(member_id).

book_id	INT	FOREIGN KEY	References Books(book_id).
borrow_date	DATE	-	Date the book was issued.
due_date	DATE	-	Deadline for returning the book.
return_date	DATE	NULLABLE	Actual date of return.

6. Table: Reviews

Stores user feedback for recommendation logic.

Column	Data Type	Constraint	Description
review_id	INT	PRIMARY KEY, AUTO_INC	Unique identifier for the review.
book_id	INT	FOREIGN KEY	References Books(book_id).
member_id	INT	FOREIGN KEY	References Members(member_id).
rating	INT	CHECK (1-5)	Numerical score from 1 to 5.
comments	TEXT	-	Textual feedback.

Relationship Summary:

- **One-to-Many:** Author → Books (An author writes many books).
- **One-to-Many:** Category → Books (A category contains many books).
- **Many-to-Many:** Members ↔ Books (via Borrow_Records and Reviews).